



Sunbeam Institute of Information Technology
Pune and Karad

Operating System Concepts and Linux Programming

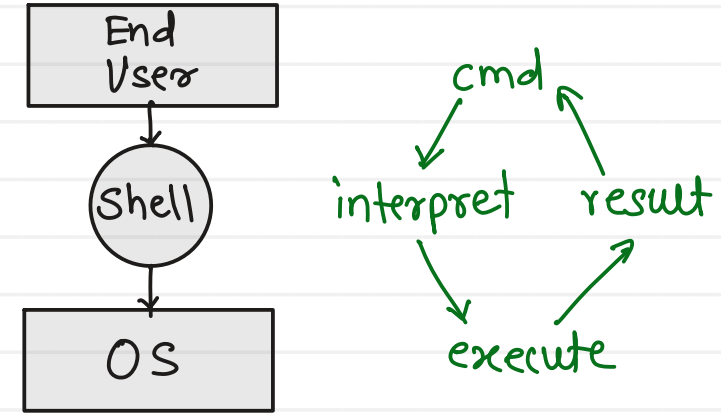
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- shell is a program present in an OS to provide interface to the user.
- shell is an intermediate betⁿ end user and operating system
- shell is a command interpreter
- in Ubuntu default shell is 'bash'

BASH - Bourne Again Shell

- echo \$SHELL - will show default shell
- chsh - to change default shell



Types of shell:

1. CLI based shell

Windows : cmd.exe, powershell.exe

Linux : bsh(sh), **bash**, csh, ksh, zsh ...

2. GUI based shell ↳ default shell

Windows : explorer.exe

Linux : xServer

└ GNOME (GNU Network Object Model Environment)

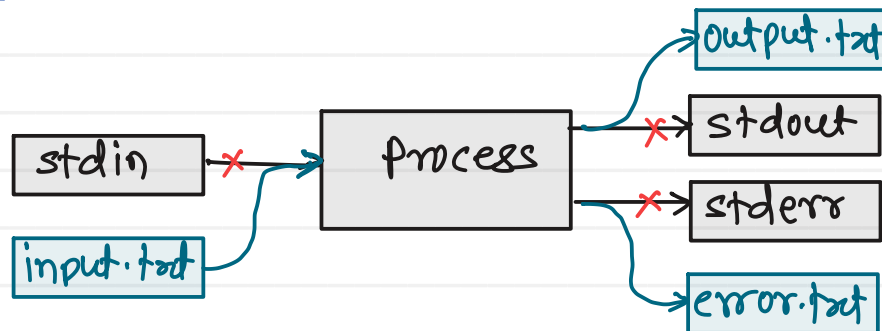
└ KDE (Kommun Desktop Environment)

* By default 3 standard file streams are opened for every process.

stdin
stdout
stderr } these standard streams are used to perform input, output and error in a process.

* default directions of input, output & error are stdin, stdout and stderr resp.

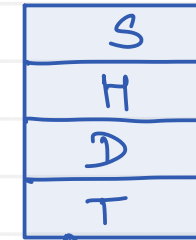
* changing the direction of input, output and error to some user defined file is called as redirection.



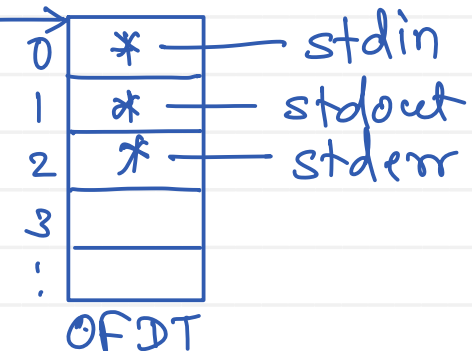
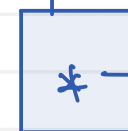
There are 3 types of redirection

1. input redirection (<)
2. output redirection (>) (<1>) (>>)
3. error redirection (2>) (<2>>)

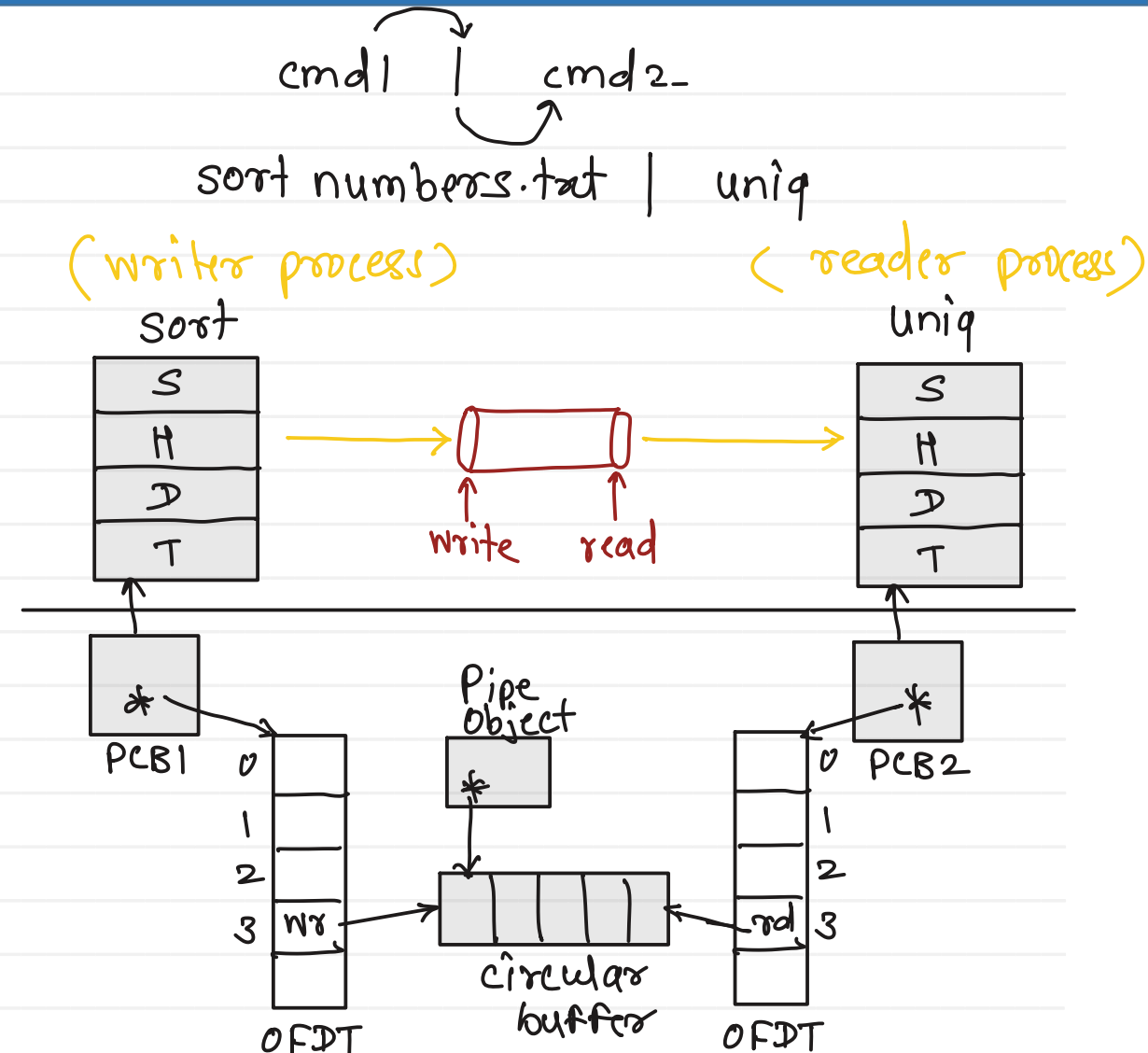
Process



PCB



- * Pipe is an IPC mechanism
- * Pipe is unidirectional IPC
- * Pipe is stream based IPC
- * Every pipe is internally a circular buffer which has two ends (read/write)
- * Both the ends of a pipe are treated as file streams
- * When write will write into full buffer, it will automatically blocked
- * When reader will read from empty buffer, it will automatically blocked



File types and permissions

File permissions/mode/access:

File type: - rW-rW-r--

1. Regular file (-)
2. Directory file (d)
3. Link file (l)
4. Socket file (s)
5. Pipe file (P)
6. Char special file (c)
7. Block special file (b)

r - read
w - write
x - execute

Levels:
user/owner
group
others

